VOICEOPS AGENT

Capstone implementation report | Status snapshot: 25 October 2026

What was built

A ZentroMart support agent that retrieves policy evidence, answers with Ollama, uses structured order and ticket tools, and supports a batch voice path through Gnani.

> Same dataset and pass checks. Step 1 is the older RAG + tools baseline; Step 2 is the updated agent.

Step 1 vs Step 2 scores

STEP	QUESTIONS	CONVERSATION RESULT / STATUS
Step 1 - completed baseline Older integrated RAG + tools agent	**25 / 52** 48.08%	8 / 96 turns passed 0 / 23 full conversations passed
Step 2 - updated agent Same RAG + tools pipeline with reliability fixes	**24 / 52** 46.15%	20 / 96 turns passed 0 / 23 full conversations passed

Step 2 change: questions -1 pass (-1.93 pp); conversation turns +12 passes (+12.50 pp); full conversations unchanged.

System flow

Customer text / audio Gnani STT for audio	**RAG retrieval** Gemini embeddings + Chroma	**Ollama + tools** Policy answer or guarded action	**Text / speech** Gnani TTS for audio

WHAT CHANGED IN STEP 2

The model providers and dataset are unchanged: Gemini is used for embeddings; Ollama remains the LLM.

Step 1 - completed baseline

The older integrated agent used RAG and structured tools. On the full evaluation it passed **25 of 52 questions** and **8 of 96 conversation turns**; no conversation passed all of its turns. Those results are the comparison point for Step 2.

Step 2 - updated agent changes

AREA	CODE CHANGE
Policy retrieval	Query rewriting uses the KB's own section names and policy vocabulary. The original customer wording remains in the search query. Relevant sections are selected for refunds, returns, warranties, verification, damage, and escalation. Current policy takes precedence over guides, help, FAQ, and marketing.
Hindi / Hinglish	Language instructions are more explicit. If the answer comes back in the wrong language, the agent retries a language-only correction while retaining the retrieval trace.
Order safety	Dates are separated from order IDs and verification digits. Changing the order clears prior verification. When the customer provides a phone suffix or pincode, tool execution uses those exact digits, even if Ollama truncates its tool argument.
Tool decisions	Clarifies missing order or payment details; handles human requests and escalation cases; only confirms a ticket after the tool reports success. Business-day calculations validate missing arguments.
Grounding	Strengthened instructions to answer from retrieved policy and verified tool results, and to say when evidence is insufficient. No canned policy answers were added.

What did not change

- The original practice dataset and its expected answers were not exposed to the agent or modified.
- Gemini embedding model, Ollama LLM, core agent architecture, and scoring annotations remain the same for the comparison runs.
- Dense Gemini retrieval remains the selected production path; retrieval failure does not switch to keyword search.

FINDINGS AND NEXT FIXES

Step 2 outcome and remaining issues

Step 2 failures to fix next

- Policy answers missed precise conditions, exceptions, timelines, or the controlling current policy section.
- Multi-turn cases depend on carrying order context, asking for verification, and withholding private data until verification succeeds.
- - Step 2 still missed lookup_order calls in 34 turns, expected clarifications in 30, and Hinglish checks in 15. These failure counts overlap.
- Ollama response time is the main run-time bottleneck: the pipeline may make separate rewrite, answer, tool-follow-up, and language-correction calls.

Measured retrieval comparison - 52 questions, top 5

RETRIEVER	RECALL@5	MRR	GOLD COVERAGE
Gemini dense	66.7%	0.675	84.6%
Gemini + BM25 RRF	70.6%	0.614	78.8%

RRF raised recall but lowered MRR and gold coverage, so dense retrieval stayed selected. No reranker was added without measured benefit; LangGraph was not justified for the current compact tool loop.

Bonus work and limits

- **29 regression tests passed** for retrieval, precedence, grounding checks, tools, verification, business-day calculations, and state isolation.
- Evaluation checkpoints preserve progress; the artifact builder emits the requested exact 13-field JSONL schema plus diagnostics and failure analysis.
- The Gnani batch voice path is implemented, but the dataset has no annotated audio set to score ASR/TTS quality.
- Current unsupported-claim checks catch numeric/date and dataset-rule violations, but do not detect every semantic hallucination.

> Step 2 result: questions 25/52 to 24/52 (-1 pass); conversation turns 8/96 to 20/96 (+12 passes). Full conversations remain 0/23. Turn-level progress did not yet produce an end-to-end passing conversation.

Annotations are evaluated after the agent responds. Original dataset files remain unchanged.